Optimising Dispatch Profitability: A Data-Driven Analysis of Temporal, Spatial, and Ride-Sharing Factors in Uber's NYC Operations

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GitHub repo with commits

1 Introduction

1.1 Background and Motivation

Every Uber ride may look like just a passenger going from A to B, but behind the scenes, profitability depends on dispatch efficiency rather than sheer trip volume. For a platform like Uber, this means making smart choices about *where, when,* and *which types of trips* to prioritise to maximise commission revenue. In a market as large and fiercely competitive as New York City, even small improvements in dispatch allocation can translate into substantial financial gains and competitive edges.

1.2 Dataset Scope and Focus

This project analyses the **TLC High Volume For-Hire Vehicle (HVFHV) Trip Record Data** for 2023–2024. We focus on 2023–2024 as our study period, as mobility patterns in New York City had stabilised from COVID-19 disruptions, providing a reliable baseline for profitability analysis. Additionally, 2023 and 2024 represent the most recent complete years of data available, allowing us to capture stable trends while ensuring that our results are based on a full dataset. (We exclude 2025, as only partial-year records were available at the time of analysis.)

This dataset covers all licensed providers, but we filter to include **Uber trips only** to examine Uber's platform-side profitability directly.

1.2.1 Profitability Measures

To capture profitability, we define two complementary response variables:

- **uber_commission = base passenger fare − driver pay** (absolute platform revenue per trip), and
- **commission per mile (y_cpm) = uber_commission / trip_miles** (relative efficiency of profit, controlling for distance).

Absolute commission identifies high-value trips, while commission per mile highlights conditions where shorter or medium trips generate proportionally greater revenue. Together, they allow us to study both the *scale* and the *efficiency* of Uber's earnings.

1.2.2 Target Audience

The target audience is **Uber dispatch operations**. Different from drivers who are primarily concerned with take-home pay, Uber, as a platform, profits through commissions. Understanding which temporal, spatial, and ride-sharing conditions increase commission revenue and efficiency enables dispatchers to refine zone pricing, rebalancing strategies, and incentives for shared rides.

1.2.3 External Datasets

To incorporate exogenous factors that shape demand and supply, we integrate:

- **Holiday calendars** from U.S. OPM (federal) and NYC OPA (city), merged into a custom CSV (`nyc_federal_holidays_2023-2024.csv`) containing date and holiday name.
- **Weather data** from NOAA's Global Historical Climatology Network (daily precipitation and temperature).
- **MTA Central Business District (CBD) Taxi Zones** to classify trips into intra-CBD, inbound, outbound, and non-CBD categories.

1.3 Methodology Overview

Our methodology proceeds in five steps:

1. Preprocess Uber HVFHV trips and derive profitability features.
2. Join external datasets and classify trips by CBD relationship.
3. Conduct exploratory analysis including distribution checks, outlier detection, and geospatial visualisation.
4. Train and evaluate two statistical models:
   - LASSO regression on **uber_commission**,
   - Random Forest on **y_cpm**.
5. Generate actionable recommendations for Uber dispatch operations.

1.4 Contribution

By combining TLC trip records with holiday, weather, and spatial data, this project provides a data-driven framework for Uber to align dispatch policies with both high revenue and high efficiency in profit generation.

2 Preprocessing

2.1 Taxi Dataset (HVFHV Uber Subset)

Removals and justifications:

- **Provider filter:** remove **non-Uber** (`hvfhs_license_num != HV0003`) so that our analysis focuses exclusively on Uber's platform profitability.
- **Missing/invalid fields:** Removed rows with nulls in essential fields (`pickup/dropoff_datetime`, `PULocationID`, `DOLocationID`, `trip_miles`, `trip_time`, `base_passenger_fare`, `driver_pay`).

- **Invalid values:** Dropped trips with non-positive (unrealistic) fares, `trip_time` $\leq$ `0`, or `trip_miles` `< 0`.
- **Temporal consistency:** Excluded trips where dropoff occurred before pickup (unrealistic).
- **Trip-level outliers:** No distance/speed capping was applied. Exploratory checks of a stratified sample revealed that virtually all trips were below 50 mph, consistent with typical NYC traffic conditions (Figure 1).
- **Commission outliers:** Negative commissions were removed, and extreme values were winsorised at the 1st and 99th percentiles to reduce the influence of rare anomalies while retaining valid high-profit trips.

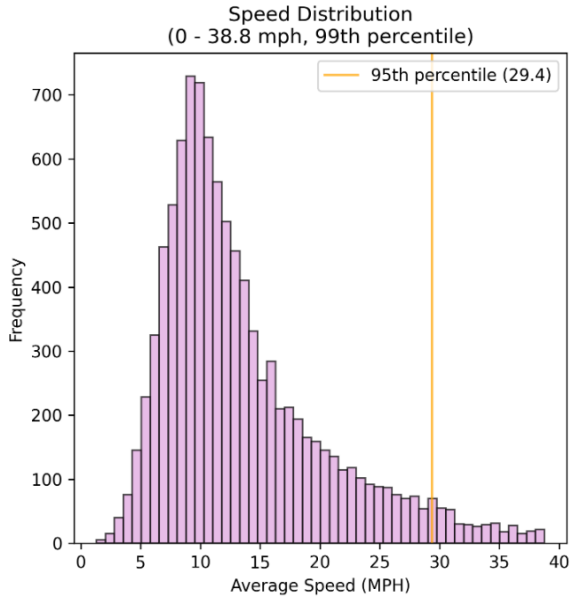


Figure 1: Speed distribution on raw data (2023), sample size = 10,000.

2.2 Dataset Shapes

Row counts at each stage are summarised in Table 1.

Stage	**2023 Rows**	**2024 Rows**	**Combined**
Raw HVFHV (all providers)	211,973,723	239,470,448	451,444,171
Uber-only filter (HV0003)	152,853,704	179,125,798	331,979,502
After quality filters	152,778,818	179,099,765	331,878,583
After sampling (none)	152,778,818	179,099,765	331,878,583
Final (after neg. comm. + wins.)	135,187,730	154,407,749	289,595,479
Retention vs raw	**63.8%**	**64.5%**	**64.1%**

Table 1: Dataset shapes across preprocessing stages.

Stage	**2023 rows**	**2024 rows**
Subsample for LASSO	64,900,884	14,792,435
Wins. on q1–98	6,355,333	14,432,214
Subsample for Random Forest	6,487,267	7,391,616
Wins. on q1–98	6,324,304	7,194,529

Table 2: Sub-sampled datasets for model training/testing with outlier removal.

Narrative: Uber-only filter removed ∼28% (2023) and ∼25% (2024) of raw records. Quality filters removed <0.05% of Uber rows. Sampling was not applied. Commission cleanup removed ∼17.6M (2023, 11.5%) and ∼24.7M (2024, 13.8%). Overall, ∼64% of trips were retained (∼290M of ∼451M).

2.3 Feature Engineering (Uber Dataset)

We engineered features to capture profitability and trip dynamics:

- **Core profitability measures:**
  - `uber_commission = base_fare -- driver_pay` (response variable).
  - `commission per mile (y_cpm) = uber_commission/trip_miles` (response variable).
- **Supporting features:** `trip_duration_minutes`, `avg_speed_mph`, `uber_take_rate`, `uber_revenue_per_minut` **driver_efficiency** = (`driver_pay` × 60)/`trip_time` (dollars per minute), and `pickup_date`.
- **Dropped fields:** Provider IDs and request metadata; pass-through charges (tolls, congestion surcharge, airport fees, sales tax, tips); deprecated metrics (`revenue_per_mile`, `revenue_per_minute`).

2.4 External Feature Integration and Engineering

To account for exogenous drivers of profitability, we integrated **spatial, temporal, and weather features**. All joins were left joins; row counts were unchanged.

3 Analysis and Geospatial Visualisation

3.1 Target Variable Distributions

Figures 2–3 show that both Uber commission and commission per mile are heavily right-skewed. Most trips generate modest returns: the median commission is only $4.44 (mean $7.08), and the median CPM is $1.68 (mean $2.33). A small fraction of high-value or very short trips drive the averages upward, indicating that profitability is dominated by a minority of rides. This confirms Uber's core challenge: balancing a high volume of low-margin trips against a small share of disproportionately profitable ones, motivating our focus on spatial, temporal, and structural drivers.

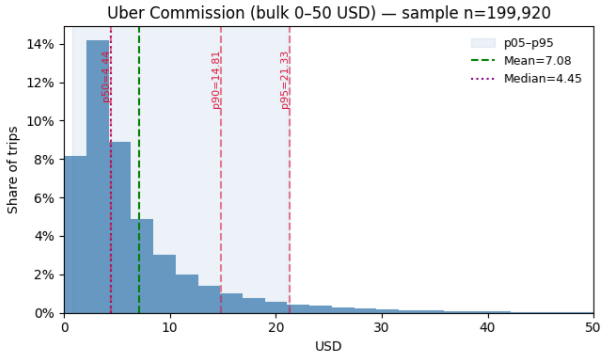


Figure 2: *

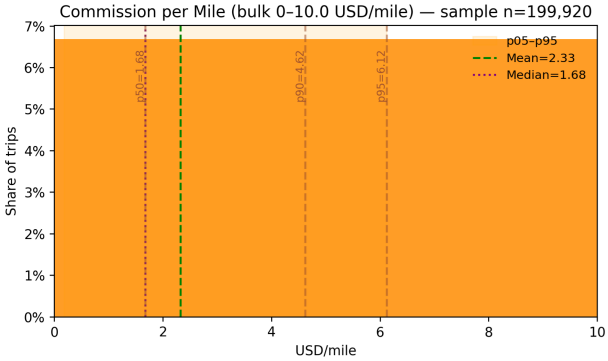


Figure 3: *

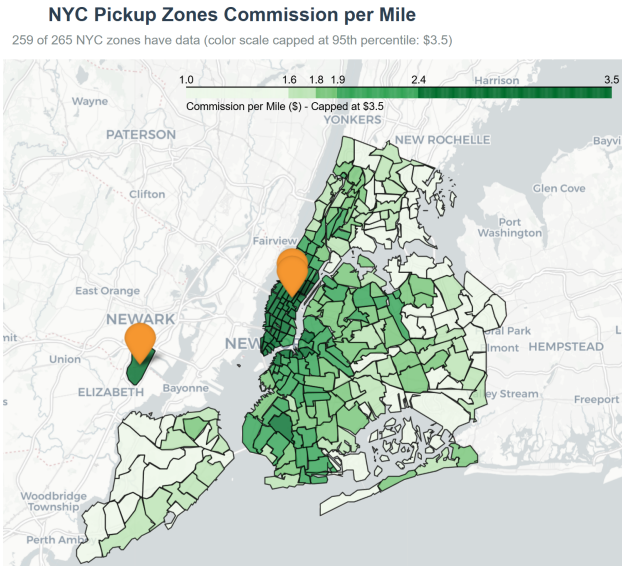
(a) Uber commission (0–50 USD)

(b) Commission per mile (0–10 USD/mile)

Figure 4: Bulk distributions of target variables. Both are right-skewed: most trips are modestly profitable, with a minority of high-value/short trips lifting the averages.

3.2 Spatial Features

Profitability clusters strongly around **airports** (JFK, LaGuardia, Newark) and **Manhattan**. Cross-borough and CBD-linked zones are above average; Staten Island and peripheral zones are consistently low, aligning with model results.



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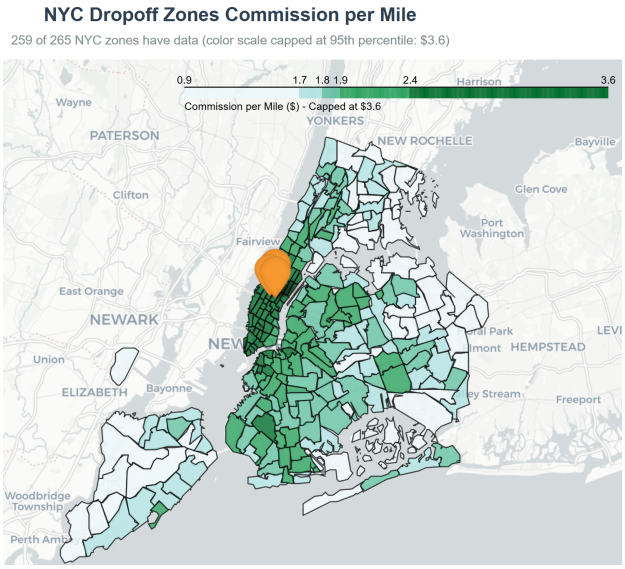

Figure 6: *

(a) Dropoff zones: CPM

(b) Pickup zones: CPM

Figure 7: Spatial profitability (commission per mile) by TLC zone. Airports and Manhattan dominate.

3.3 Ride-sharing Features

Ride-sharing effects vary strongly by distance. For very short trips (0–2 miles), shared rides achieve higher efficiency ($5.19/mile vs. $3.72), as fixed base fares are spread over minimal distance. Beyond 2 miles, shared rides underperform: at 10–20 miles they return $0.56/mile vs. $1.00 for non-shared, and over 20 miles efficiency falls to $0.41/mile. In absolute commission, non-shared trips dominate. Sharing increases profitability only for ultra-short trips and erodes efficiency for medium/long trips.

3.4 Weather Features

Weather conditions exert little influence on profitability. Average commission per mile is largely unchanged across temperature, precipitation, wind, and snow categories; minor fluctuations are within noise and do not persist in modelling (weather signals are shrunk to zero). Dispatch profitability appears structurally and geographically driven rather than weather-driven.

3.5 Temporal–Structural Interactions

Intra-CBD trips are most efficient per mile (early-morning peak), but non-CBD trips dominate volume, so aggregate profitability is driven by high-volume non-/cross-borough rides.

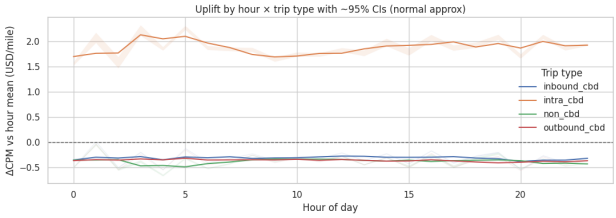

Figure 8: *

(a) Hourly uplift in CPM by trip type

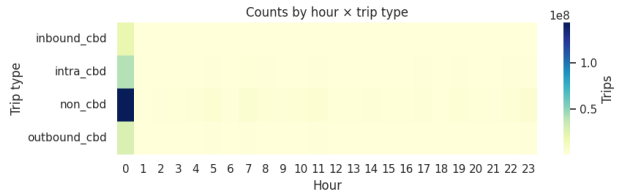

Figure 9: *

(b) Trip counts by hour × trip type

Figure 10: Temporal–structural patterns. (a) Intra-CBD rides have the highest per-mile profitability; (b) non-CBD rides dominate demand across all hours.

4 Modelling

4.1 Lasso Regression (Uber Commission)

Lasso regression is a shrinkage method that reduces variance by introducing a small amount of bias, thereby improving predictive performance in high-dimensional settings. Unlike Ridge regression, Lasso produces sparse solutions by shrinking many coefficients exactly to zero, enabling variable selection and interpretability [1].

Since our target is a numerical response variable (**uber_commission**) with a very high number of predictors (65,608), we fit a Generalised Linear Regression model with an ℓ_1 penalty (Lasso). This is particularly suitable given the large feature space created through one-hot encoding of categorical variables. The model is trained on **50% subsamples of 2023 trips** for training and **10% subsamples of 2024 trips** for testing, with high-cardinality features hashed and low-cardinality features one-hot encoded.

The Lasso optimisation problem has the standard form:

$$\hat{\beta} = \arg\min_{\beta} \left\{ \|y - X\beta\|_2^2 + \lambda \|\beta\|_1 \right\},$$

with regularisation parameter set to $\lambda = 0.1$. We assume additive effects with no explicit interaction terms.

4.2 Random Forest (Uber Commission per Mile)

Random Forests are an ensemble learning method that combine many decision trees, each trained on a bootstrapped sample of the data with random feature sub-sampling. By aggregating across trees, Random Forests reduce variance compared to a single tree, offering strong predictive performance and robustness to overfitting [2]. The trade-off is reduced interpretability relative to linear models.

This flexibility makes Random Forests well-suited for our analysis of **commission per mile (y_cpm)**, as they can capture non-linear relationships and interactions between spatial, temporal, and trip-structural features. Our model is trained on **5% subsamples of 2023–24 data** for both training and testing, with **numTrees = 120** and **maxDepth = 12**.

4.3 Model Performance and Inference

Lasso Regression. Despite modest predictive accuracy ($R^2 = 0.27$), the Lasso highlights interpretable structural drivers of Uber commission. Table 4 shows that **airports dominate positively**, especially LaGuardia (+5.94 combined effect), followed by JFK and Newark. Within NYC, **Manhattan pickups**

Table 3: Model performance on test sets.

Model	**RMSE**	**MAE**	**R^2**
Lasso Regression (Commission)	4.52	3.08	0.27
Random Forest (Commission per Mile)	1.48	1.09	0.28

Table 4: Aggregated view of top-50 Lasso coefficients (post-hoc grouping).

Group	**Signed sum**	**Count**
Airports (DO)	+8.39	7
Airports (PU)	+2.42	2
Airports (PU+DO combined)	+10.81	9
Manhattan (PU)	+1.85	6
Brooklyn (PU)	+1.49	7
Manhattan (DO)	+0.68	5
Bronx (PU)	-0.24	1
Bronx (DO)	-0.20	1
Queens (PU)	-0.08	1

Table 5: Aggregated Random Forest feature importances.

Group	**Top features**	**Combined importance**
Spatial zones	DO_Zone (24.3%), PU_Zone (23.9%)	48.2%
Trip structure	is_cross_borough (16.1%), trip_cbd_type (14.7%)	30.8%
Service zones	PU_service_zone (4.2%), DO_service_zone (3.4%)	7.6%
Temporal	pickup_month (2.7%), ym_idx (2.5%)	5.2%
Other (minor)	weather, holiday, rush hour, etc.	$< 2\%$

and parts of Brooklyn contribute positively, while Bronx and Queens have small negative signals. Trip structure also matters: intra-CBD trips are strongly negative (–1.77), whereas outbound-CBD trips are mildly positive (+0.35). Numeric features (duration, distance, speed) contribute linearly; temporal effects are weaker; weather effects are negligible (penalised to zero).

Random Forest. The Random Forest achieves slightly higher explanatory power ($R^2 = 0.28$, RMSE = 1.48) and captures non-linear interactions. Table 5 shows that **spatial zones** explain nearly half of commission-per-mile variation, while **trip structure** (cross-borough and CBD involvement) adds ∼31%. Temporal factors contribute moderately; external features (holidays, weather) are negligible (<2%).

Comparison. Lasso quantifies marginal effects on total commission (airports, Manhattan, trip length). Random Forest highlights non-linear structural and temporal interactions (cross-borough, CBD). Both support the EDA finding that geography and structure dominate profitability.

Implication for Dispatch Strategy. Uber can improve platform profitability by prioritising **airports, Manhattan, and cross-borough trips**, especially during high-demand periods, and by de-emphasising frequent short intra-CBD trips.

5 Discussion: Limitations and Future Work

Limitations We used **subsampled datasets** due to computational limits (50%/10% for Lasso train/test; 5%/5% for Random Forest). **Hyperparameters were not exhaustively tuned** (e.g., λ

heuristic; 120 trees, depth 12), so predictive accuracy is not the primary focus; interpretability for dispatch is.

Future Work: Uber Shuttle Services Our findings highlight airport trips as most profitable. This aligns with Uber's shared airport shuttle service: launched to **LaGuardia in Oct 2024** and expanded to **JFK in Apr 2025** [3, 4]. These fixed-route, pre-booked shuttles create a new *shared-ride* product extending airport-linked profitability. Future work should integrate shuttle booking/ridership to evaluate commission efficiency in shared airport services.

6 Recommendations

1. Prioritise airport trips. Allocate drivers toward LaGuardia, JFK, and Newark during peak flight waves; tune dispatch to favour airport-linked matches.

2. Encourage cross-borough and CBD-linked rides. Use surge/bonuses to favour trips spanning boroughs or linking outer boroughs with Manhattan.

3. Deprioritise short intra-CBD trips in peaks. During rush hours/weekends, prefer longer or cross-borough rides over short intra-CBD demand.

4. Align incentives with temporal demand cycles. Use weekend/evening incentives and rider promos to match high-demand periods.

5. Minimise weight on low-impact external factors. Weather/holiday signals showed negligible impact; keep focus on geography, structure, and demand cycles.

7 Conclusion

We analysed Uber's dispatch profitability in NYC using Lasso and Random Forest models, supported by EDA. **Airport trips, Manhattan-linked rides, and cross-borough journeys** are consistently most profitable; frequent short intra-CBD trips yield lower efficiency. Temporal patterns matter secondarily; weather/holidays are negligible. We recommend prioritising airport-linked and cross-borough trips, aligning incentives with demand cycles, and strategically managing intra-CBD dispatch. Uber's new shared airport shuttles reinforce airports' central role and offer a promising direction for future shared-ride analysis.

Acknowledgements

I would like to acknowledge the use of AI-assisted tools during the preparation of this project. These tools supported me in tasks such as drafting and refining LaTeX formatting, structuring sections of the report, and debugging or optimising code snippets. They also assisted in improving the clarity and conciseness of explanations and ensuring figures and tables were consistently integrated. Importantly, all modelling decisions, data processing steps, and interpretations of results were carried out and validated by me, and I take full responsibility for the analysis and conclusions presented in this report.

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[2] University of Melbourne. *Lecture 13: Classifier Combination and Random Forests*. MAST30034 Applied Data Science course material. 2025.

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[4] Dennis Schaal. *Uber Expands Shuttle Service to JFK Airport*. Mar. 2025. URL: `https://skift.com/2025/03/27/uber-expands-shuttle-service-to-jfk` (visited on 08/31/2025).